DUST on the K=32 code-carried model: the 500-step $\sigma\times$ `dust_lr_scale` grid

Research note

2026-10-22; Section 7 added 2026-10-24.

Single runs, seed 1337; the encoder arm and the baseline trained on new_york (8× RTX 5090), the backprop arm on georgia (8× A100), the $K = 8$ runs of Section 7 on paris (8× RTX 5090).

1 Setup

The model is the K=32 code-carried hard Top-32 model of `docs/kj-vs-hard-topk-code-residual.tex` (`configs/cr_kj/ma_cr_hard_k32_pnorm.yaml`): 8 blocks of width 768, abs-TopK over $N = 1536$ features at `residual_out` of every block, the code carried between blocks, post-norm, magnitude–direction (MD) decoupling, batch 96×512 tokens, a 20k-step cosine schedule with 500 warmup steps.

Every DUST run (`configs/cr_dust/hp500/`, method in `docs/dust-topk.md`) adds the DUST bottleneck optimization: the $J = 480$ pool receives the noise, population 128 (16 draws per bottleneck), schedule $[1, 1]$ (every other optimizer step a DUST step), $\gamma = 0$, gradient scale 1, a separate optimizer for the DUST steps sharing the MD parameterization, and no gradient clipping. The grid is `dust_target` $\in \{\text{encoder}, \text{backprop}\} \times \sigma \in \{0.03, 0.1, 0.3, 0.9\} \times$ `dust_lr_scale` $\in \{1, 10, 30, 100, 300\}$. Each run trains 500 steps of the unchanged 20k-step schedule, so the window lies inside the LR warmup ($1.2 \cdot 10^{-6}$ to $6 \cdot 10^{-4}$; the DUST steps use that value times `dust_lr_scale`), with validation every 50 steps on 40 batches of 24 windows.

The baseline is the same model trained with the same settings and no DUST steps (`ny_hp500_hard_k32`); the original madrid run of the model (gradient clipping at 1.0) is quoted for reference.

2 Encoder target

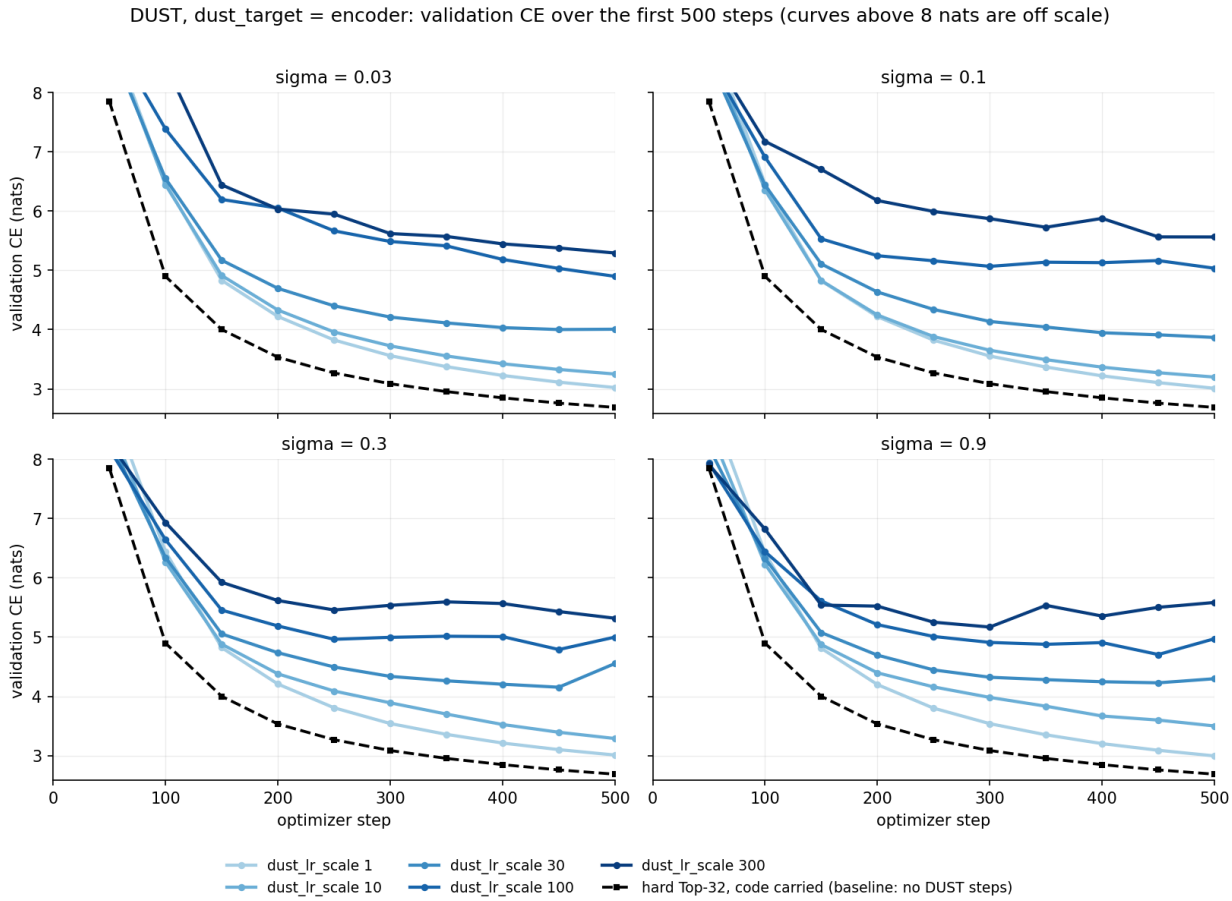


Figure 1: `dust_target = encoder`: validation CE over the first 500 steps, one panel per σ, one curve per `dust_lr_scale` (one hue, light to dark with the multiplier); dashed black, the hard Top-32 baseline. Curves above 8 nats are off scale.

Validation CE at step 500 (bold: the best cell):

σ \ `dust_lr_scale`	1	10	30	100	300
0.03	3.0220	3.2500	4.0055	4.8971	5.2906
0.1	3.0096	3.1989	3.8673	5.0334	5.5611
0.3	3.0101	3.2891	4.5535	4.9989	5.3168
0.9	**2.9957**	3.5009	4.2972	4.9715	5.5825
hard Top-32 baseline (no DUST steps, same settings)	2.6875				
hard Top-32, the original madrid run (clipping 1.0)	2.6846				

3 Backprop target

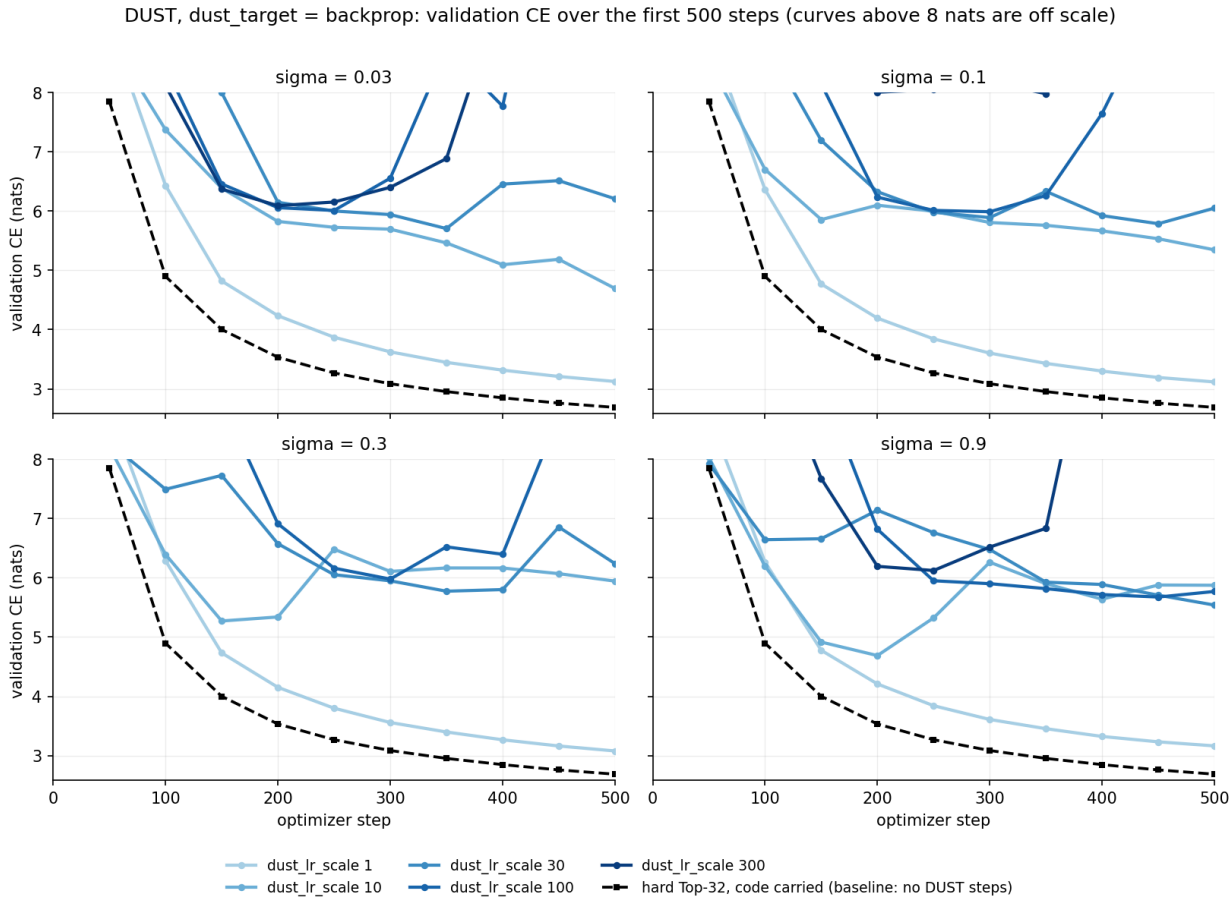


Figure 2: `dust_target = backprop`: the same layout. Curves above 8 nats are off scale.

Validation CE at step 500:

σ \ `dust_lr_scale`	1	10	30	100	300
0.03	3.1246	4.6930	6.2055	13.0487	9.7345
0.1	3.1169	5.3437	6.0478	8.8164	9.9774
0.3	**3.0794**	5.9428	6.2382	11.9248	9.6194
0.9	3.1646	5.8732	5.5386	5.7687	9.2661
hard Top-32 baseline (no DUST steps, same settings)	2.6875				
hard Top-32, the original madrid run (clipping 1.0)	2.6846				

4 Results

No cell reaches the baseline. In both arms the validation CE at step 500 rises monotonically with `dust_lr_scale`, and the multiplier 1 is the best value at every σ: 2.996 to 3.022 in the encoder arm, 0.31 to 0.33 nats above the baseline, and 3.08 to 3.16 in the backprop arm. The dependence on σ is weak at the multiplier 1 (a spread of 0.03 nats in the encoder arm and 0.09 in the backprop arm) and does not change the ordering by the multiplier anywhere.

The two targets differ in how the larger multipliers fail. In the encoder arm the curves keep descending at every setting, only more slowly and from a higher level as the multiplier grows; at 100 and 300 they plateau near 5 to 5.6 nats within the window. In the backprop arm, where the

estimate is backpropagated to every parameter before the gate inputs, every multiplier from 10 on stalls near 5 to 6 nats, and 100 and 300 diverge at every σ (final values of 5.8 to 13 nats), with the pre-activation scale growing by orders of magnitude within the first tens of steps.

Within the first 500 steps the DUST steps therefore cost against hard steps at every setting of the grid, least when the DUST learning rate is the schedule's own, and the encoder target is less harmful than the backprop target at every σ and multiplier.

5 Sigma sweep at population 1024

`dust_lr_scale` 1, population 1024 (128 draws per bottleneck, eight times the grid's), both targets, the four σ values, otherwise the grid's settings; all eight runs on new_york.

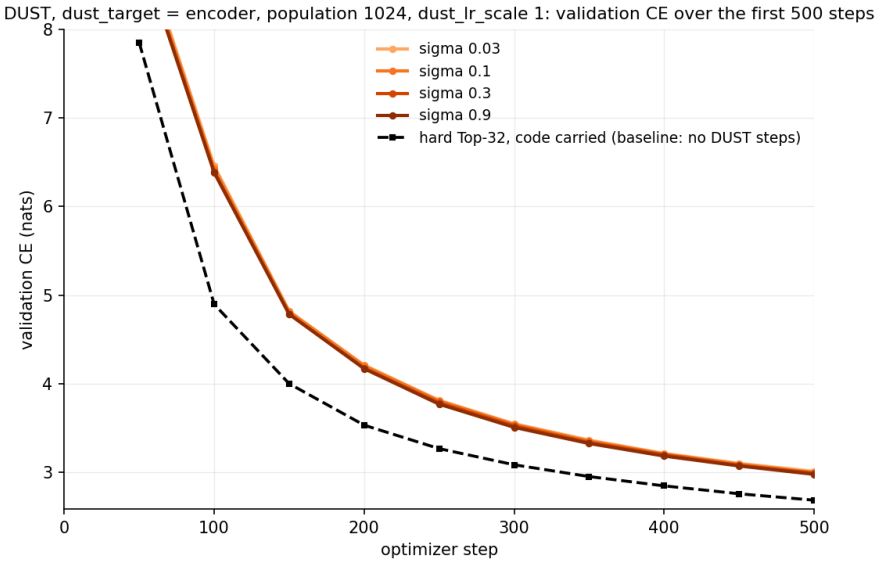


Figure 3: Population 1024, `dust_target = encoder`: validation CE over the first 500 steps, one curve per σ (one hue, light to dark with σ); dashed black, the hard Top-32 baseline.

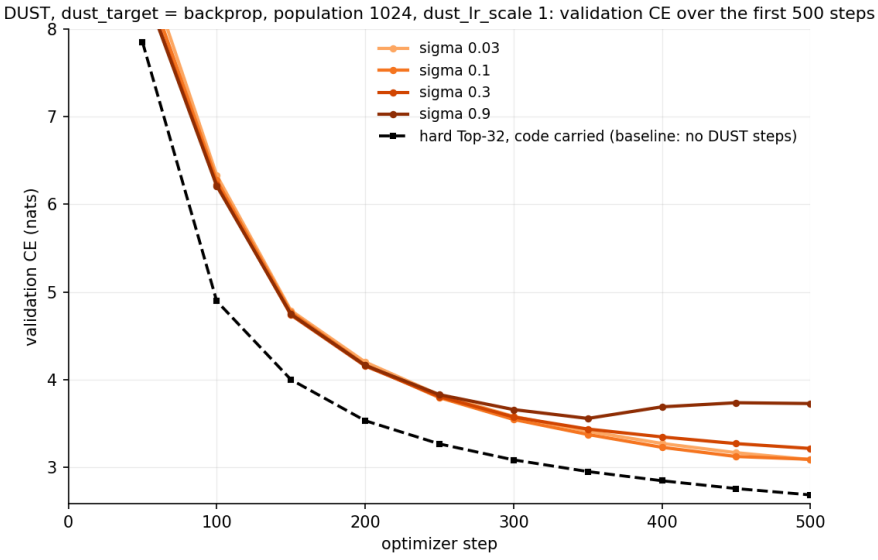


Figure 4: Population 1024, `dust_target = backprop`: the same layout.

Validation CE at step 500 (population 128 at `dust_lr_scale` 1 in parentheses):

`dust_target` \ σ	0.03	0.1	0.3	0.9
encoder	3.0117 (3.0220)	2.9975 (3.0096)	2.9923 (3.0101)	**2.9790** (2.9957)
backprop	3.0896 (3.1246)	3.0930 (3.1169)	3.2166 (3.0794)	3.7286 (3.1646)
hard Top-32 baseline (no DUST steps)	2.6875			
hard Top-32, madrid run (clipping 1.0)	2.6846			

Eight times the population lowers the encoder arm's CE at step 500 by 0.010 to 0.018 nats at every σ, and the four curves are indistinguishable until the last 100 steps, where they order by σ with the largest best. The backprop arm improves by 0.02 to 0.04 at $\sigma = 0.03$ and 0.1 and worsens at 0.3 and 0.9, where the curves turn upward after step 350 ($\sigma = 0.9$ ends at 3.73). The gap to the baseline, 0.29 nats in the encoder arm at best, is therefore not a matter of the estimate's variance at this population: the estimate's variance is eight times smaller and the gap moved by a few hundredths.

6 Without MD decoupling

The hard Top-32 run and the population-1024 σ sweep (`dust_lr_scale` 1, both targets) repeated with `model.decouple = false` and `model.md_init = true`: the initialization is the same, only the optimizer changes, from the decoupled one (directions on fixed-norm spheres, learnable gains, no weight decay) to ordinary AdamW with weight decay 0.1, the convention of `docs/md-init-vs-decoupling.tex`. The hard run and the $\gamma = 0$ sweep on georgia, the $\gamma = 0.99$ sweep on new_york; checkpoints at steps 250 and 500.

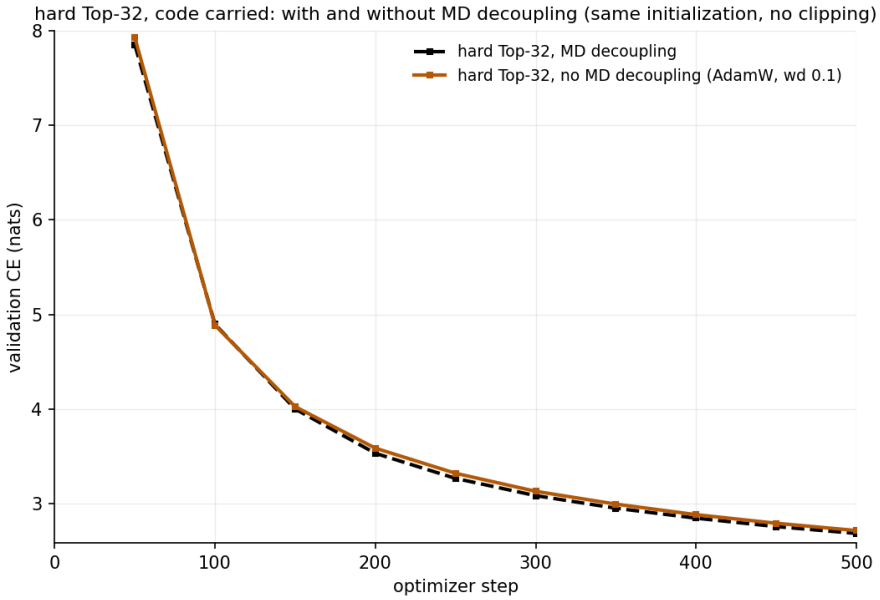


Figure 5: Hard Top-32, code carried, with (dashed black) and without (solid) MD decoupling: same initialization, no clipping, validation every 50 steps.

run	validation CE at step 500
hard Top-32, MD decoupling	2.6875
hard Top-32, no MD decoupling (AdamW, weight decay 0.1)	2.7181

The two hard runs coincide to step 150 and separate by 0.03 to 0.06 nats from step 200 on; at step 500 the decoupled optimizer is 0.031 nats ahead.

$\gamma = 0$

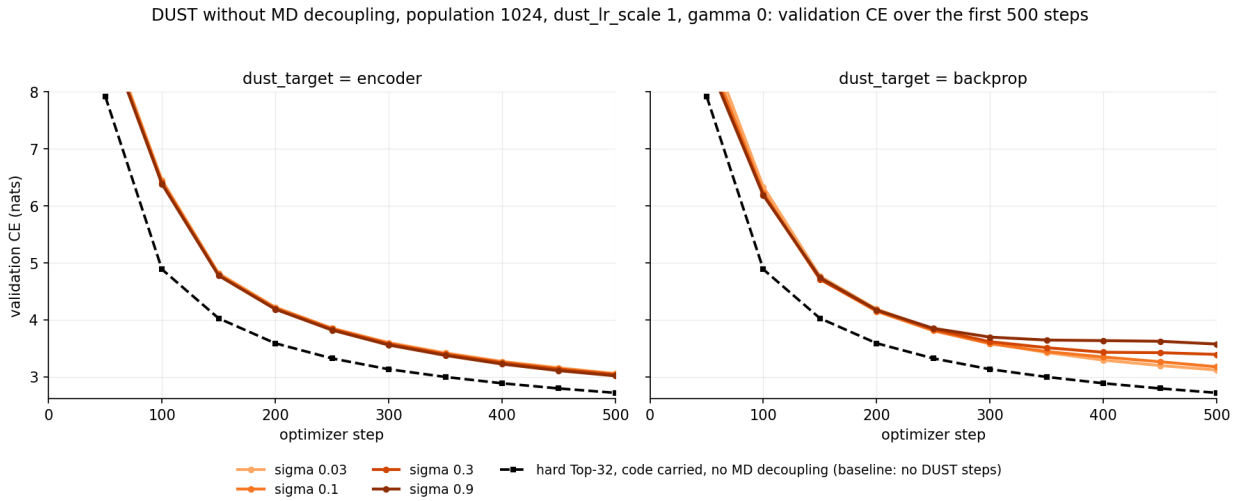


Figure 6: DUST without MD decoupling, population 1024, `dust_lr_scale` 1, $\gamma = 0$: encoder target left, backprop target right, one curve per σ; dashed black, the hard Top-32 run without decoupling.

Validation CE at step 500, without MD decoupling (with MD, from the sweep of Section 5, in parentheses):

`dust_target` \ σ	0.03	0.1	0.3	0.9
encoder	3.0558 (3.0117)	3.0492 (2.9975)	3.0369 (2.9923)	**3.0163** (2.9790)
backprop	3.1192 (3.0896)	3.1748 (3.0930)	3.3926 (3.2166)	3.5744 (3.7286)
hard Top-32, no MD (no DUST steps)	2.7181			

Removing the decoupling moves the DUST runs by about as much as it moves the baseline: the encoder arm ends 0.30 to 0.34 nats above the no-MD hard run (0.29 to 0.32 above the MD hard run with decoupling), ordered by σ with the largest best and the four curves indistinguishable to step 400. The backprop arm ends 0.40 to 0.86 nats above it; its curves separate after step 250 and flatten for $\sigma = 0.3$ and 0.9, which end at 3.39 and 3.57 (with decoupling, $\sigma = 0.9$ turned upward to 3.73). The gap between DUST and the hard model is therefore not a property of the decoupled optimizer.

$\gamma = 0.99$

The same eight runs with `dust_gamma` $= 0.99$: a token's noise is rewarded with its own loss change plus the discounted loss changes of the later tokens.

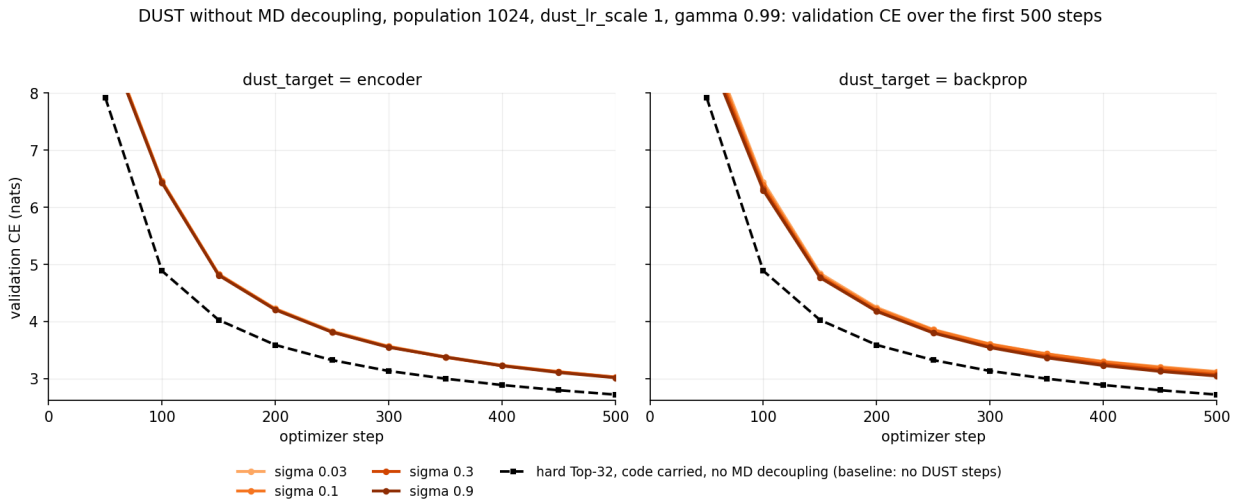


Figure 7: As above, $\gamma = 0.99$.

Validation CE at step 500 ($\gamma = 0$ in parentheses):

`dust_target` \ σ	0.03	0.1	0.3	0.9
encoder	3.0248 (3.0558)	3.0193 (3.0492)	3.0241 (3.0369)	**3.0139** (3.0163)
backprop	3.1229 (3.1192)	3.1082 (3.1748)	3.0665 (3.3926)	3.0437 (3.5744)
hard Top-32, no MD (no DUST steps)	2.7181			

The discounted credit changes the backprop arm most: the flattening at $\sigma = 0.3$ and 0.9 disappears (3.39 to 3.07 and 3.57 to 3.04), the arm is now ordered by σ with the largest best, and it ends within 0.03 of the encoder arm. The encoder arm gains 0.01 to 0.03 nats at $\sigma = 0.03$ to 0.3 and nothing at 0.9, with its four curves again on top of one another. Both arms stay 0.30 to 0.40 nats above the hard run. The per-draw signal-to-noise ratio of the estimate (`dust/snr`, defined in `docs/dust-topk.md`) is 0.088 to 0.091 in every one of these sixteen cells at step 500, independent of σ, target and γ, so at 128 draws per bottleneck the population estimate has a per-token signal-to-noise ratio of about 1.

7 $K = 8$ without LR warmup

The same model with $K = 8$ and $J = N - K = 1528$, so the whole pool receives the noise (`configs/cr_dust/k8/`). The hard Top-8 run keeps every setting of the hp500 baseline (MD decoupling, no clipping, validation every 50 steps) except `train.warmup_steps` $= 0$: the learning rate starts at its peak $6 \cdot 10^{-4}$ and follows the 20k-step cosine from there, so the 500-step window no longer lies inside the warmup. The four DUST cells use that schedule and the grid's settings (schedule $[1, 1]$, $\gamma = 0$, `dust_lr_scale` 1, a separate optimizer), with population 2048 given to a single bottleneck per DUST step (`dust_layers_per_step` $= 1$; the eight bottlenecks in turn, so each is perturbed every sixteenth optimizer step with 2048 draws): `dust_target` $\in \{\text{encoder}, \text{backprop}\} \times \sigma \in \{0.2, 1.0\}$, 500 steps each. A DUST step took 235 s on a 5090 (16 h per cell). The hard Top-8 run with the 500-step warmup, the convention of every earlier hard run, is quoted for reference; both hard runs completed the full 20k steps.

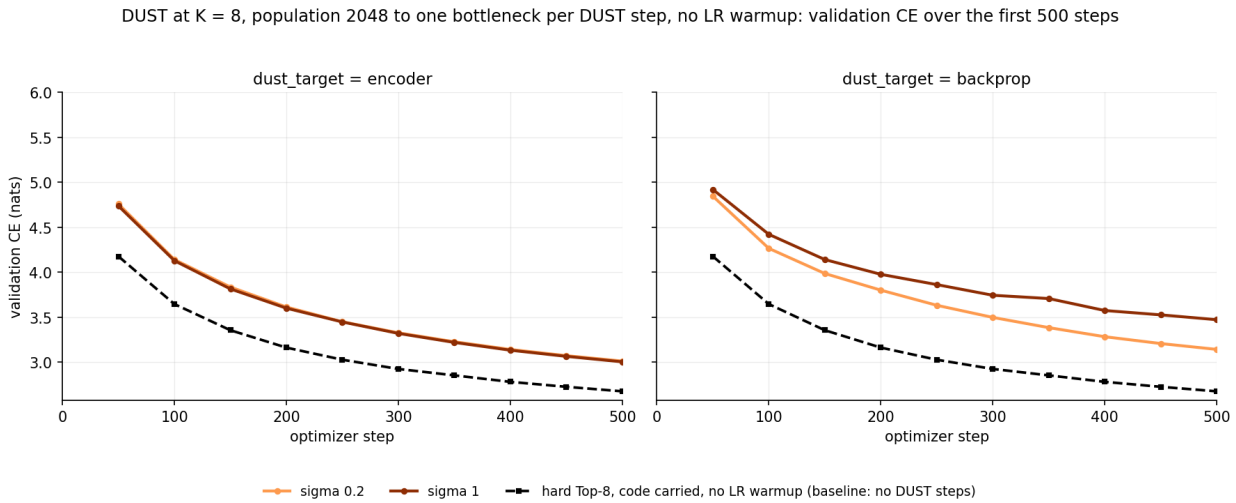


Figure 8: $K = 8$, no LR warmup: encoder target left, backprop target right, one curve per σ; dashed black, the hard Top-8 run without warmup.

Validation CE at step 500:

`dust_target` \ σ	0.2	1.0
encoder	3.0137	**3.0053**
backprop	3.1437	3.4738
hard Top-8, no LR warmup (no DUST steps, same settings)	2.6777	
hard Top-8, with the 500-step LR warmup	2.9127	

The encoder cells end 0.33 to 0.34 nats above the hard run without warmup, the gap of the $K = 32$ cells (0.29 to 0.34 nats in Sections 2, 5 and 6), with the two σ curves within 0.02 nats of each other at every validation. The gap narrows over the window, from 0.57 at step 50 to 0.33 at step 500, and both cells also end above the hard run with warmup (2.913 at step 500). The backprop cells end 0.47 ($\sigma = 0.2$) and 0.80 nats ($\sigma = 1.0$) above the hard run; at $\sigma = 1.0$ the gap grows from 0.74 at step 50 to 0.85 at step 350 and the curve flattens between steps 300 and 350, the behaviour of the $\gamma = 0$ backprop cells at $\sigma = 0.9$ in Sections 5 and 6. The per-draw signal-to-noise ratio of the estimate is 0.023 to 0.026 in every cell at step 500, at the pure-noise level $1/\sqrt{2048} = 0.022$ of 2048 draws, as the $K = 32$ cells sat at $1/\sqrt{128}$: giving the whole population to one bottleneck leaves the per-token signal-to-noise ratio of the population estimate near 1. Over the full schedule the hard run without warmup ends at 1.7586 and the one with warmup at 1.7923.

8 Runs on Drive

- Encoder arm, new_york: `gdrive:weight-sparsity/runs_new_york/ny_hp500_enc_sg<s>_lr<l>` for `<s>` $\in$ {`0p03`, `0p1`, `0p3`, `0p9`} and `<l>` $\in$ {1, 10, 30, 100, 300}, twenty runs.
- Baseline, new_york: `gdrive:weight-sparsity/runs_new_york/ny_hp500_hard_k32`.
- Population-1024 sweep, new_york: `gdrive:weight-sparsity/runs_new_york/ny_hp500_<a>_p1024_sg<s>_lr1` for `<a>` $\in$ {`enc`, `bp`}, eight runs.
- Backprop arm, georgia: `gdrive:weight-sparsity/runs_georgia/ga_hp500_bp_sg<s>_lr<l>`, the same twenty cells.

- Without MD decoupling, georgia: `gdrive:weight-sparsity/runs_georgia/ga_hp500_hard_k32_nomd` and `gdrive:weight-sparsity/runs_georgia/ga_hp500_<a>_p1024_nomd_sg<s>_lr1`, eight runs.
- Without MD decoupling, $\gamma = 0.99$, new_york: `gdrive:weight-sparsity/runs_new_york/ny_hp500_<a>_p1024_nomd_g099_sg<s>_lr1`, eight runs.
- $K = 8$ (Section 7), paris: `gdrive:weight-sparsity/runs_paris/pa_k8_hard` (warmup, full 20k steps), `gdrive:weight-sparsity/runs_paris/pa_k8_hard_nowarm` (no warmup, full 20k steps) and `gdrive:weight-sparsity/runs_paris/pa_k8_dust_<a>_sg<s>_nowarm` for `<a>` $\in$ {`enc`, `bp`} and `<s>` $\in$ {`0p2`, `1p0`}, four runs with checkpoints at steps 250 and 500; the hard runs carry the checkpoints that reached Drive before pruning (every 250 steps, the last two kept), including step 20000.
- Original hard Top-32 run (reference value only): `gdrive:weight-sparsity/runs_madrid/ma_cr_hard_k32_pnorm`.

Figures and tables: `scripts/plot_dust_hp500.py` over the `metrics.jsonl` of these runs. The Markdown version of this note, `docs/dust/hp500-sigma-lr.md`, lists every run name in full.